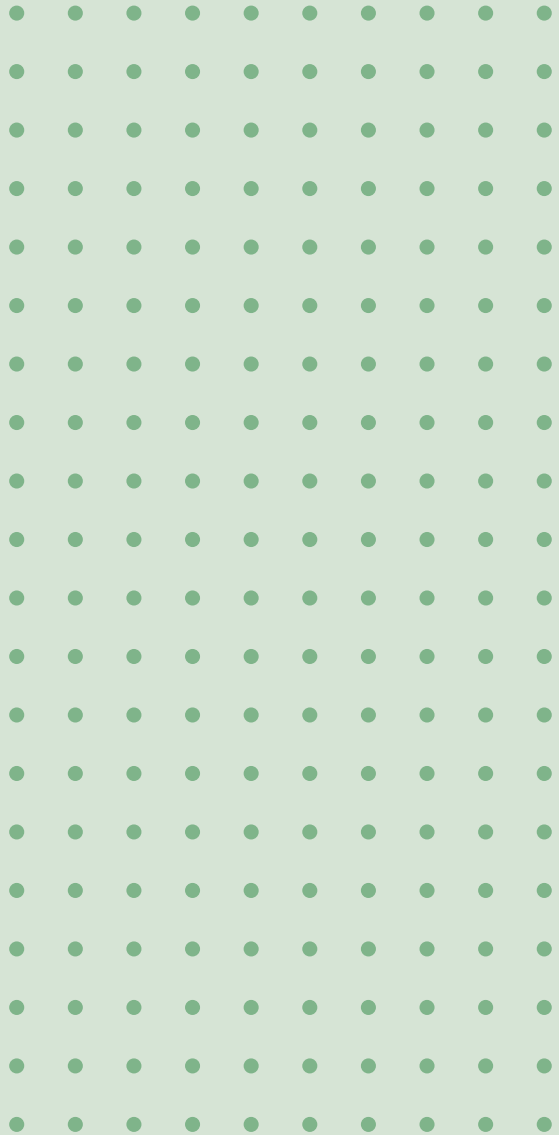




Golden Paths and Guardrails

Building Norgespris on Kubernetes

May 2026

Agenda

- Who are we?
- Elhub before Norgespris
- Elhub's Tech Stack
- Enabling the Golden Path
- Authentication & Authorization
- Monitoring: the PLG stack
- Elhub after Norgespris

Introduction

Michael Akinde

- Development Group Lead, Elhub 2016 - *
- Software Architect, Met.no 2006-2016
- Database Specialist, SMHI 2002-2007
- Assistant Lecturer, Aalborg Uni., 2001-2002
- Vis. Researcher, AT&T Labs, 2000

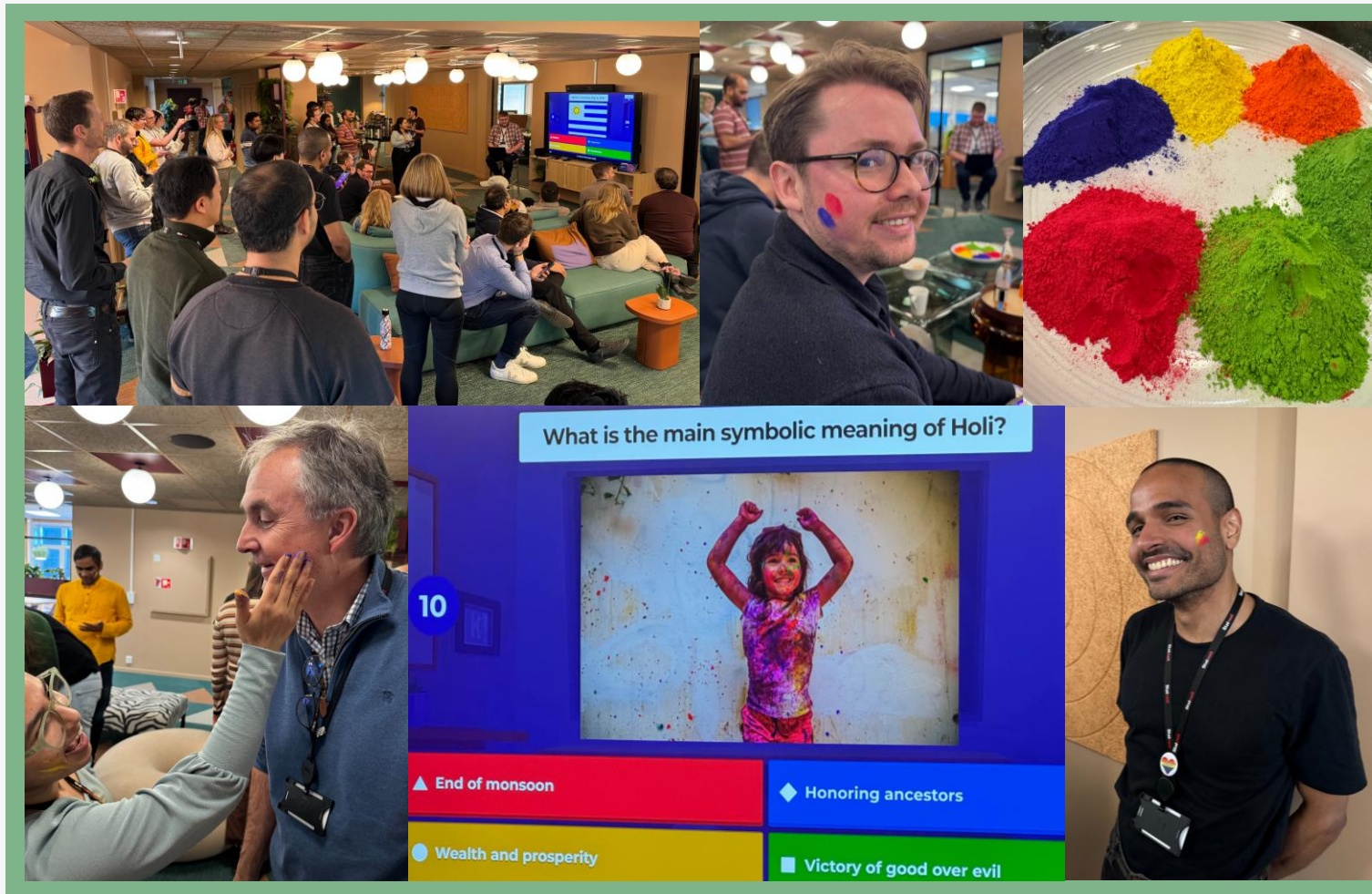
Johanna Grøvdal Schaanning

- Software Engineer, Elhub 2021 - *
(test-data -> devxp -> devxp+kubernetes)



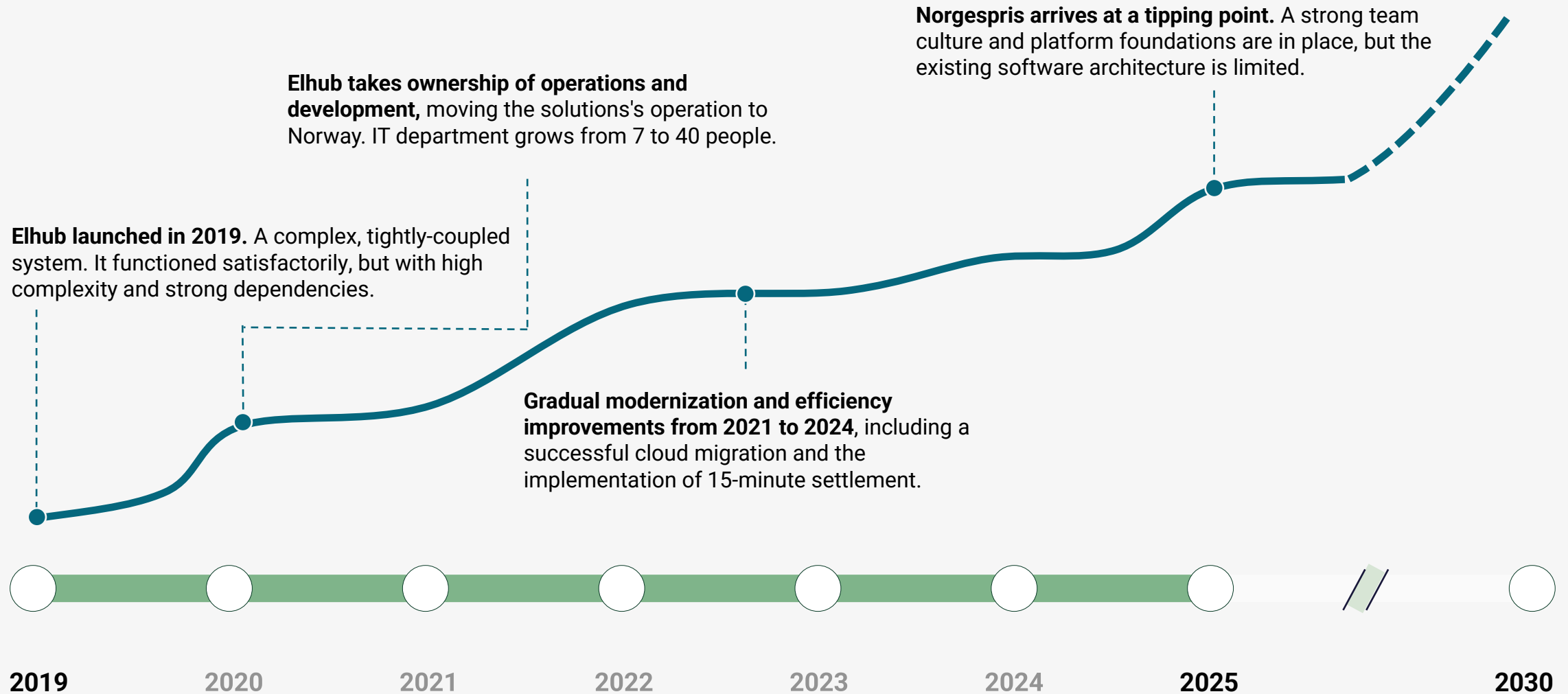
Statnett's subsidiary IT company with 80 employees

- Development, operations, and domain knowledge
- Digital value chain for collecting and distributing data in the Norwegian power market
- Regulated by the government and funded through the Elhub fee determined by RME



Recently worked on: **Norgespris** 😊

Elhub's Journey



Norgespris

- Tight implementation deadline
- Elhub was given a central role:
 - The electricity customer ordering portal
 - Distribution of data between market participants
 - Central registry of who has placed an order
 - Providing the regulator with data for monitoring and oversight
- Strong collaboration across the industry
- Key Success Factors for Elhub:
 - Agile organization structured around value-stream teams
 - **Modernization of the technical platform**
 - Experience in building and operating solutions for the entire industry

Aftenposten.no 31. januar 2025:

Aftenposten

Nyheter

Oslo

Meninger

A-magasinet

Vink

E-avis

Bli abonnent

Økonomi | Strømprisene

Nye strømgrep: Sjekk hva «norgespris» betyr for deg

Arbeiderpartiet innfører fastprisordning på strøm. Slik påvirker dette lommeboken din.



Folk vil merke strømtiltakene på lommeboken, lover Støre. Foto: Marius Helge Larsen / NTB

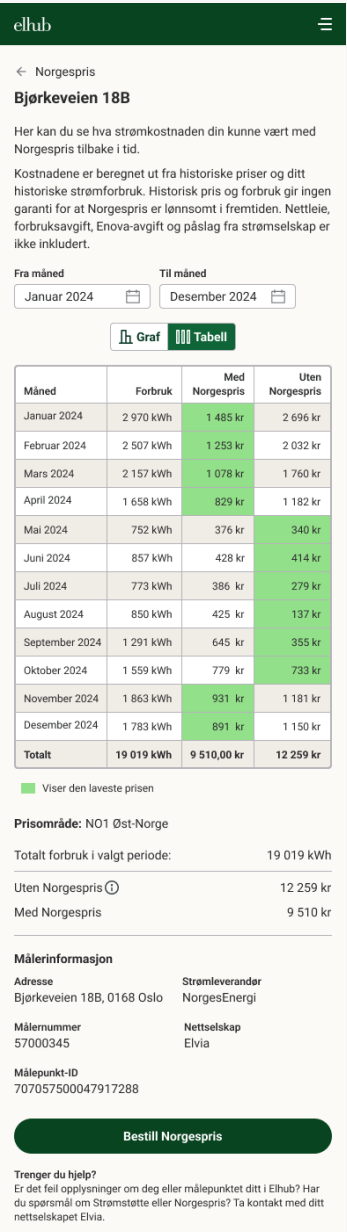
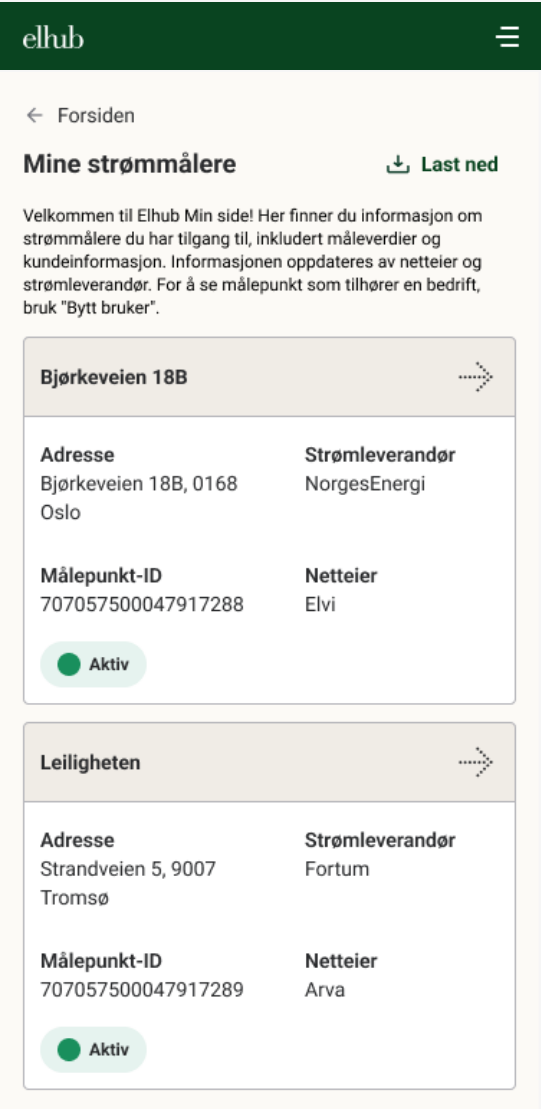
Andreas Brattåker Støyva Journalist

Preben Brækstad Journalist

Fredrik Thoresen Redaksjonell utvikler

Publisert: 31.01.2025 11:30 | Oppdatert: 31.01.2025 15:54

Norgespris: Just a button...



Norgespris: Just a button...

- How many people?

Gjøvik

Folkeavstemningen kollapset etter få minutter: – Flaut og pinlig



Foto: Per Skjønberg og skjermdump

Av [Per Skjønberg](#), [Tobias Brøste Båkind](#), [Erik Børresen](#) og [Thomas Amlien](#)

Oppdatert: 22.04.26 12:22
Publisert: 22.04.26 10:23

Kopier

Enkelte fikk gitt sin stemme om felles kommune, så streiket systemet i en halvtime.

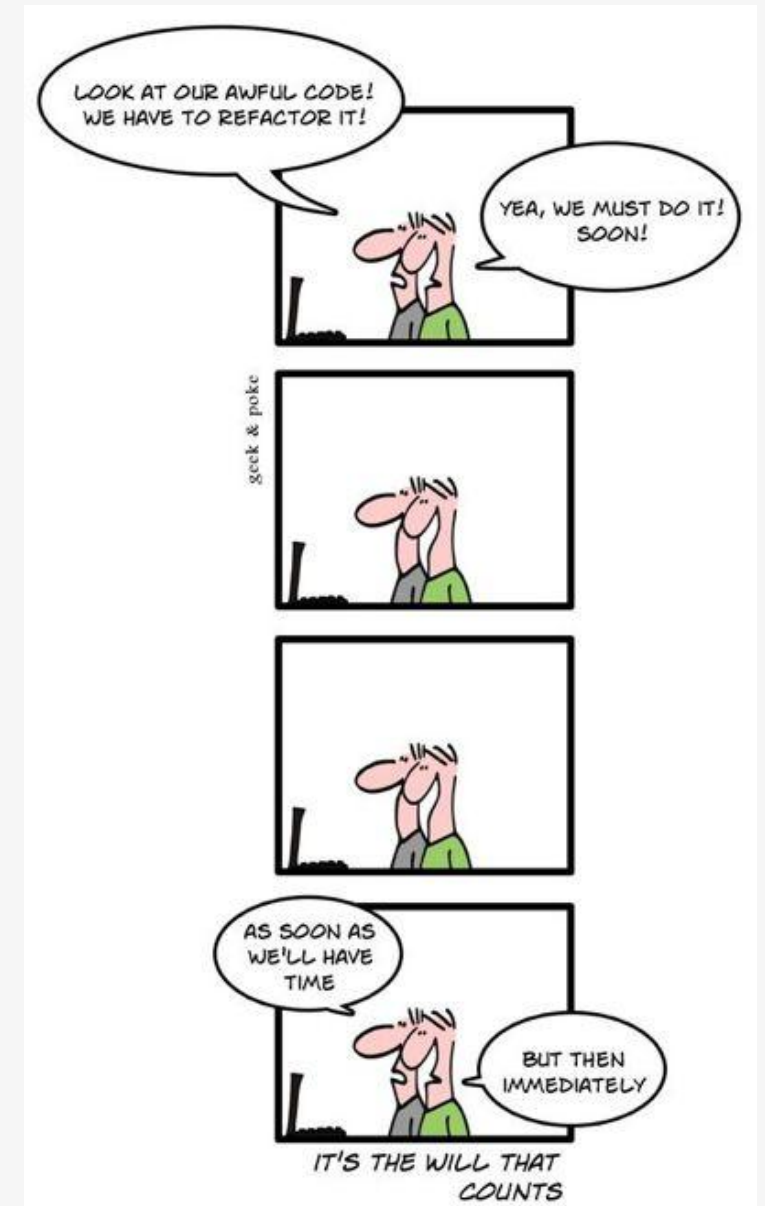
Klokka 10 onsdag ble det åpnet for å stemme i folkeavstemningen om

Norgespris: Just a button...

- Tight coupling into Elhub's core platform
 - Long deployment cycles, downtime
- Dependency on legacy Oracle components
 - Running on virtual machines (not SaaS)
- Not dimensioned for high user loads
 - ~50,000 lifetime users up to 2025
- Also, the old UI sucked

Norgespris: Solution

- Rewrite (almost?) everything
 - We were planning to do so anyway...
- Migrate to **Kubernetes and containers**
- Implement a modern authentication and authorization stack
 - Open Policy Agent (OPA) as a dedicated auth service
- Modernize the observability stack



Our 'cloud native' stack

 kubernetes CNCF GRADUATED	 cilium CNCF GRADUATED	 KEDA CNCF GRADUATED	 CERT MANAGER CNCF GRADUATED	 Grafana	 PostgreSQL		
				 Grafana loki	 HUBBLE	 aqua trivy	
 argo CNCF GRADUATED	 HELM CNCF GRADUATED	 Prometheus CNCF GRADUATED	 etcd CNCF GRADUATED	 TeamCity	 Liquibase	 HashiCorp Vault	 Jfrog Artifactory
				 kafka	 Gradle	 unleash	 HashiCorp Terraform
 KEYCLOAK CNCF INCUBATING	 Open Policy Agent CNCF GRADUATED	 cri-o CNCF GRADUATED	 OpenCost CNCF INCUBATING	 CoreDNS CNCF GRADUATED	 Kyverno CNCF GRADUATED		

What do we do 'differently'?



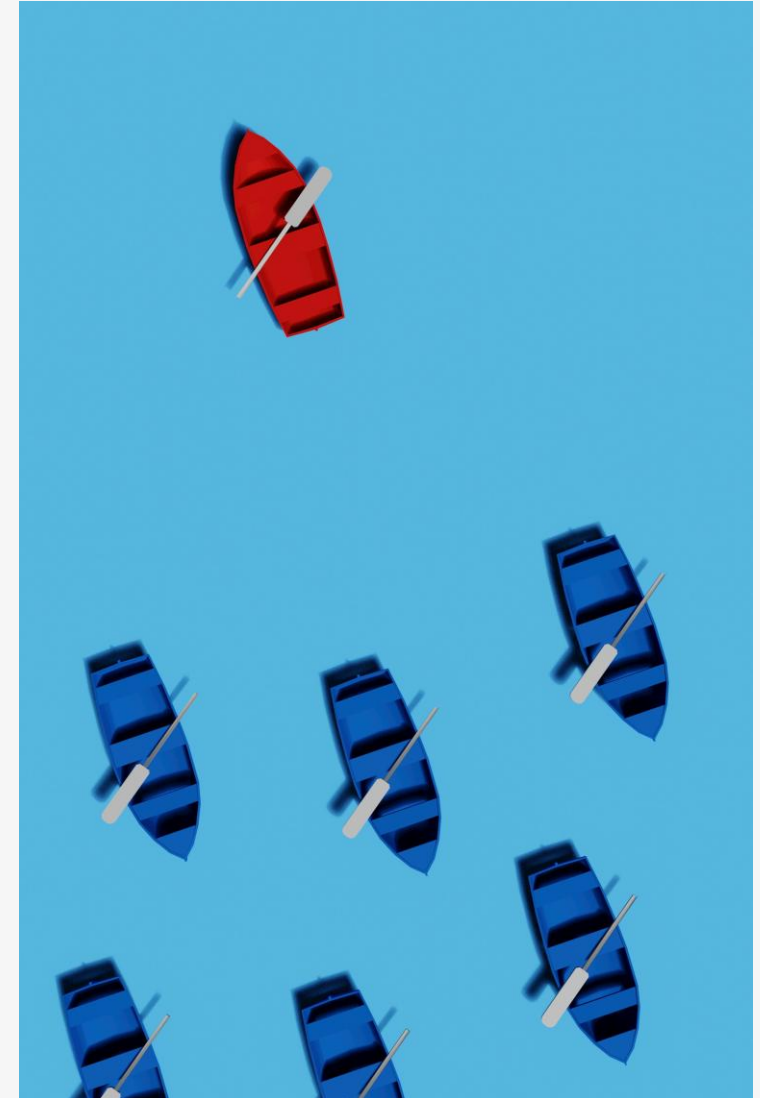
TeamCity with custom DSL



Helm with `Kustomize-like structure`



→ (different=better)



TeamCity and custom DSL

 Already somewhat niche with ~7% market share

 Kotlin DSL > UI setup

 Custom DSL on top of JetBrains' DSL



'Vanilla' Kotlin DSL

```
project { this: Project
    val gradleVerify = buildType { this: BuildType
        // job that runs build and test +. redacted for brevity
    }
    buildType { this: BuildType
        id("GradleJib".toId())
        name = "Gradle Jib"
        description = "Publish Docker image using Gradle Jib"

        steps { this: BuildSteps
            script { this: ScriptBuildStep
                name = "Login to Registry"
                scriptContent = """
                    echo 'env.TOKEN%' | docker login -u %env.USERNAME% --password-stdin %env.REGISTRY%
                """
            }
            gradle { this: GradleBuildStep
                name = "Publish Image"
                tasks = "jib"
                gradleParams = """
                    -Djib.from.image=alpine:latest
                    -Djib.to.tags=latest,%env.BUILD_TAG%
                    -Djib.to.image=docker.registry.org.cloud/teamname/image-name
                """.trimIndent()
            }
            script { this: ScriptBuildStep
                name = "Publish Tag"
                scriptContent = """
                    <script that creates and pushes tag to git repository>
                    <call to that script>
                """
            }
        }

        dependencies { this: Dependencies
            snapshot( buildType = gradleVerify) { this: SnapshotDependency
                onDependencyFailure = FailureAction.FAIL_TO_START
            }
        }
    }
}
```

Our custom DSL

elhub

```
elhubProject(Group.DEVXP, name = "some-repository-name") { this: ProjectScope
    pipeline { this: PipelineScope
        sequential { this: PipelineScope
            gradleVerify()
            gradleJib()
        }
    }
}
```

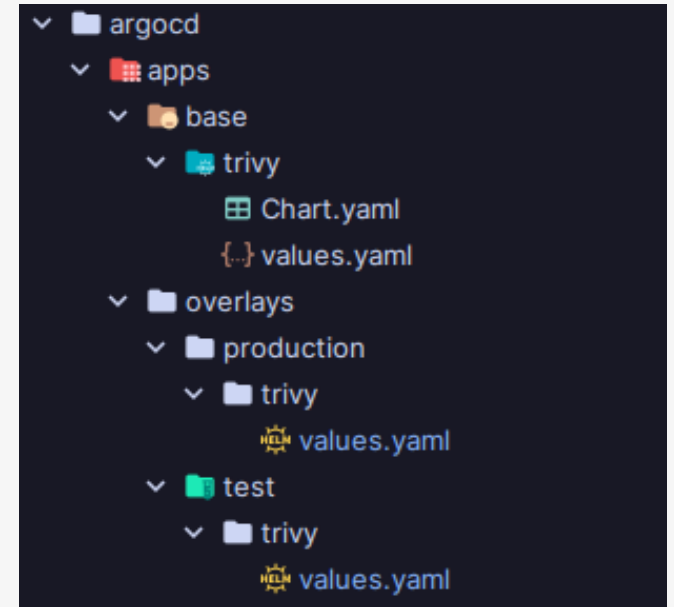
```
elhubProject(Group.DEVXP, name = "some-repository-name") { this: ProjectScope
    pipeline { this: PipelineScope
        sequential { this: PipelineScope
            gradleVerify { this: ...GradleVerifySettings.Builder
                javaVersion = JavaVersion.VERSION_17
                modules = listOf("module1", "module2")
            }
            gradleJib { this: GradleJibSettings.Builder
                javaVersion = JavaVersion.VERSION_17
                registrySettings = { this: RegistrySettings.Builder
                    repository = "elhub/my-project"
                }
            }
        }
    }
}
```

Helm with 'Kustomize-like' structure

 base/overlays

+ helm-merge plugin (for ArgoCD)

🏛️ Devs only modify Chart.yaml values.yaml
(and potentially templates) files



```
apiVersion: argoproj.io/v1alpha1
kind: ConfigManagementPlugin
metadata:
  name: helm-merge-plugin
spec:
  version: v1.0
  init:
    command: ["/plugin/scripts/merge-and-render.sh"]
    args: ["init"]
  generate:
    command: ["/plugin/scripts/merge-and-render.sh"]
    args: ["generate"]
```

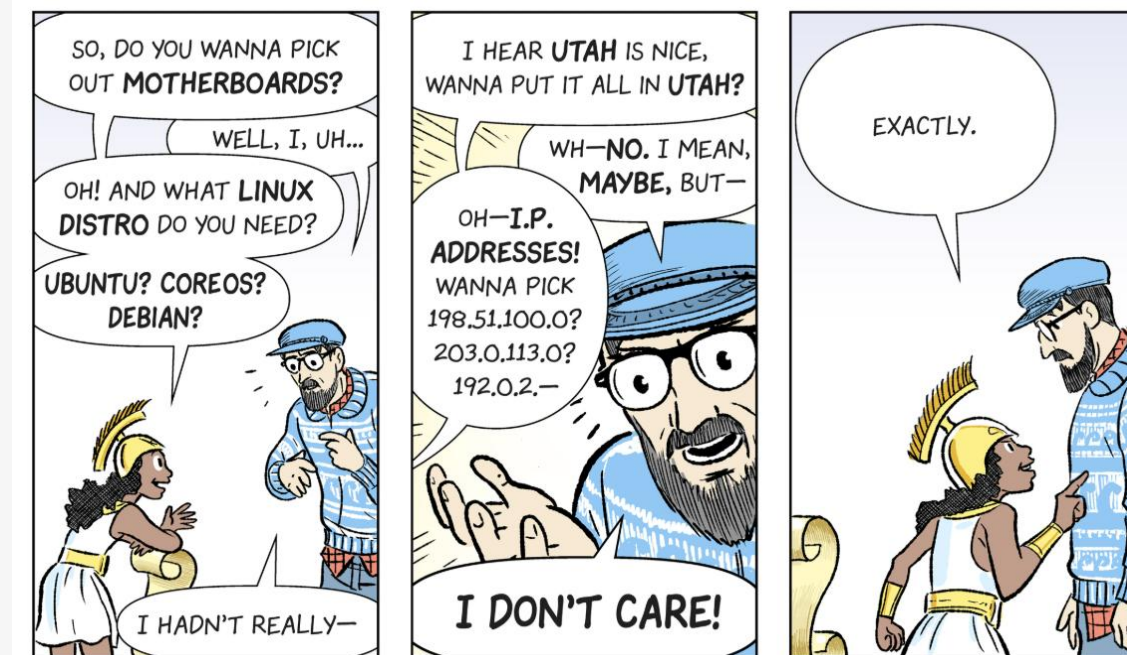

'Golden-path'-ing

 Repository templates for api creation

 TeamCity pipelines for 'standardized' images

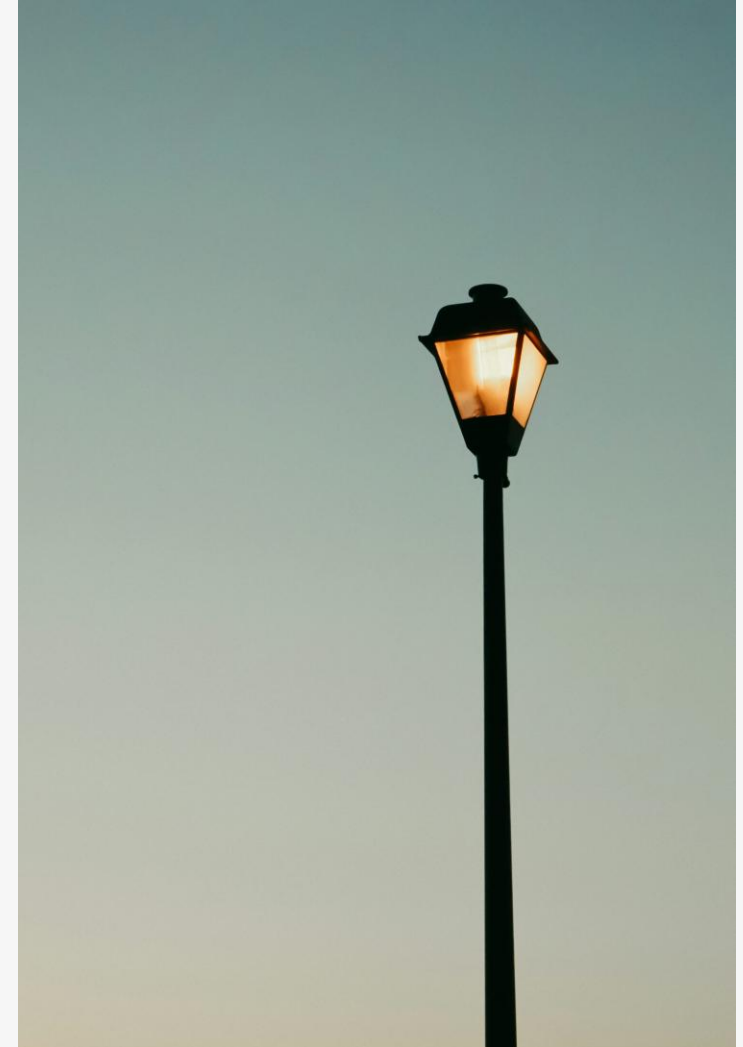
 Base Helm Charts for app configuration

-  Application
-  CronJob
-  File Storage
-  HttpRoute



Benefits

-  Testable jobs/charts
-  Abstracts away 'how'-logic
-  Unified application/pipeline structure
-  Easy 'Golden Path' for devs
-  Light 'gatekeeping' of bad practice

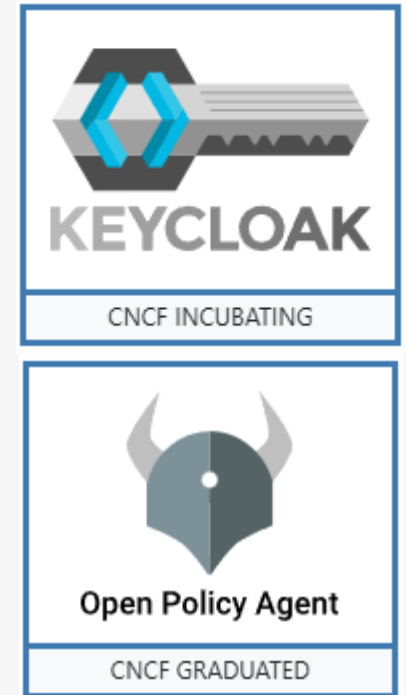


Authentication & Authorization

- Authentication and authorization embedded in UI and service code
- Tight coupling between frontend, backend and identity logic
- IDporten deeply integrated and hard to upgrade
- Certificate-based service-to-service communication
- Strong dependencies on legacy Oracle-based components (not services)

Authentication & Authorization: The Shift

- Backend-For-Frontend (BFF) layer
 - Split between UI – Identity/Access – Domain
- ID-porten for user authentication
- Maskinporten for organization authentication
- Keycloak as the central authentication provider
- Open Policy Agent (OPA) as a centralized authorization service



Authentication & Authorization: Experiences

- Cleaner service code and simpler UIs
- Consistent authorization across services
- Looser coupling
- But:
 - Data synchronization
 - Need to guard against business logic sneaking into authorization
 - Risk of Auth services becoming the new SPOF

Things we learned



JVM tuning can be hard



Managing many clusters can be demanding



Monitoring can be resource-intensive

JVM tuning in k8s

Heap space percentage

- 256MB limit = ~127MB heap
- 512MB limit = ~127MB heap 🤔
- <256MB -> 50% | 256-512MB -> ~127MB | >512MB -> 25%

JVM memory allocation and OOMKills

```
-XX:MaxRamPercentage 75.0
```

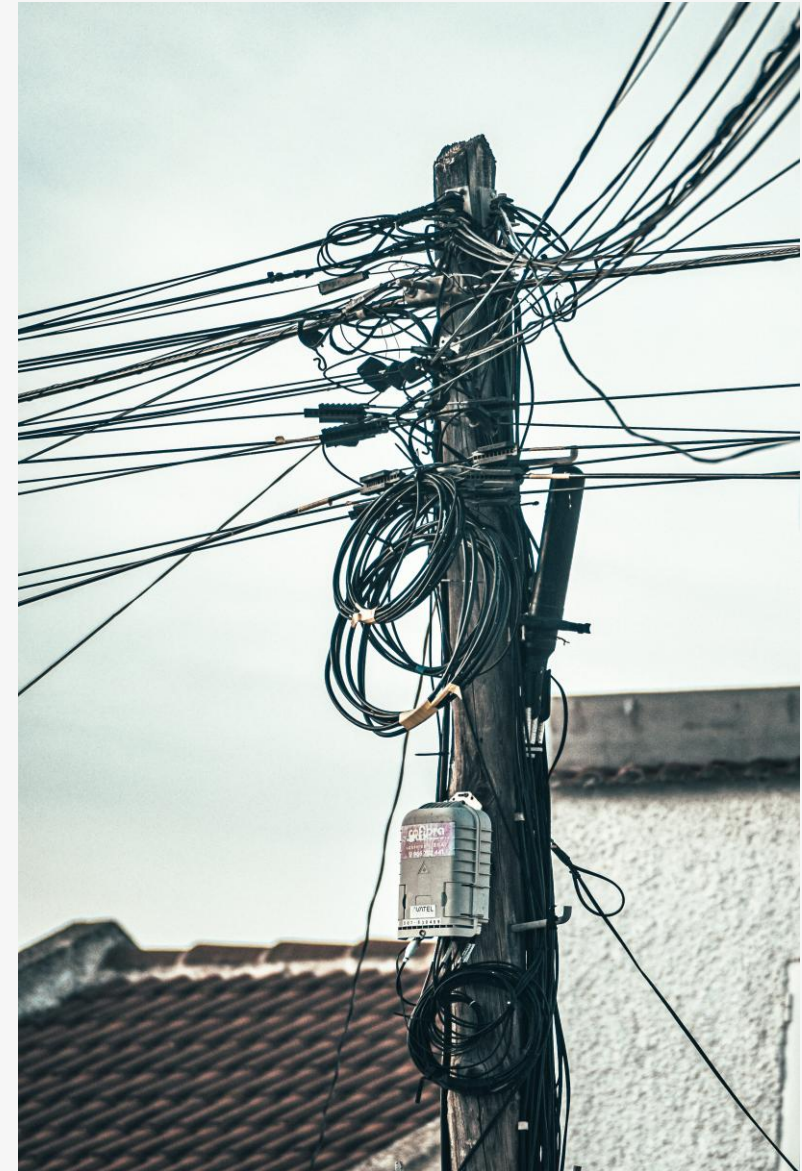
```
resources:  
  limits:  
    memory: 2Gi  
  requests:  
    memory: 1Gi
```



Setting limits = requests helps avoid this but can also be wasteful

Cluster maintenance

- Ported elhub's 'legacy' environments
 - production, staging, trial
 - infra, infra-test
 - test*
- 12 34 9 instances of most apps
11 instances of ArgoCD|Cilium|etc.
- ↓ Working on consolidating into fewer clusters

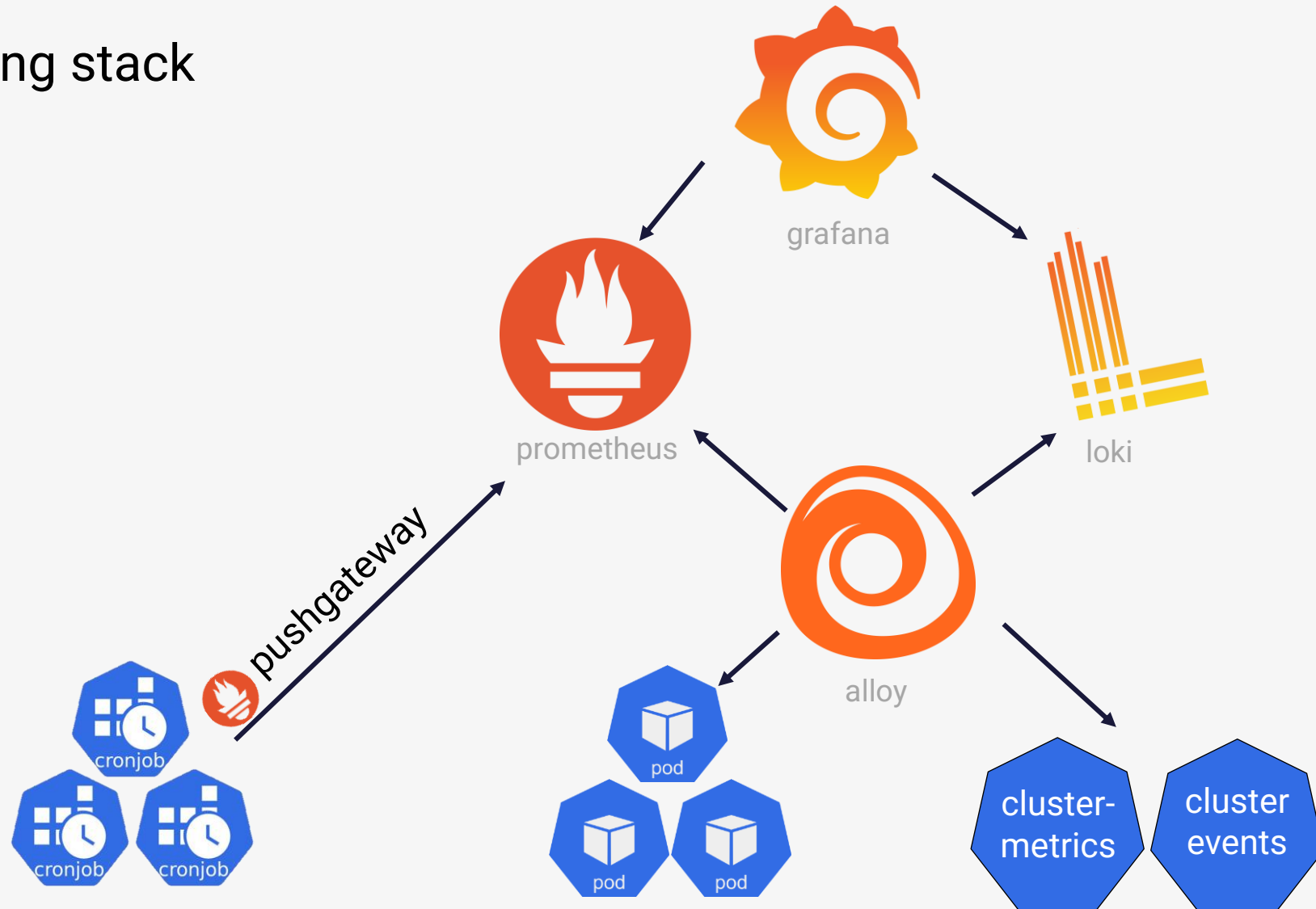


Monitoring can be resource intensive

🔧 Managing the monitoring stack

- Setup
- Tuning
- Dashboard creation

📊 Resource usage



Launch Day



Generally very smooth

- No application downtime
- Overdimensioned resources



Grafana issues

- Large dashboard with many views
- ~1 week interval
- 10+ active users
- 5 sec default refresh

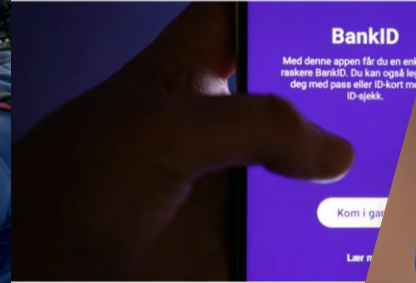
Someone said our metrics
aren't working!

Better check Grafana!



tv2.no

elhub



Trøbbel



Antall minside inlogginger over tid



▶ 01:29





- First Day
 - 380,000 users
 - 334,000 contracts
 - 19 peak logins/sec
 - 50ms response
- Today
 - 1.44 million users
 - 1.61 million contracts

1. Valgomat
2. Charlie Kirk
3. Ed Gein
4. Ozzy Osbourne
5. Labubu
6. Norge Italia
7. Bodø Glimt
- 8. Elhub**
9. Odd Nerdrum
10. Stig Millehaugen

Topp 10
søketrender i 2025

